

# **Proposed Internship Idea Document**

## **Idea Title: Design and Development Smart Crutch for Injury Rehabilitation**

Name: Nikhil Sai Kenguri  
Department: Artificial Intelligence and Machine Learning (CSE)  
College: Kalasalingam Academy of Research and Education  
Email ID: sainikhil73865@gmail.com  
LinkedIn Profile: www.linkedin.com/in/nikhilsai-kenguri-0b0976322

## **1. Introduction**

After lower-limb injuries or surgeries, patients are advised to use crutches to reduce pressure on the injured leg. But in real situations, most patients do not know how much weight they are actually putting on the leg while walking. This can either slow down recovery or even cause further injury. In hospitals, doctors guide patients properly, but at home there is no monitoring. Many patients walk with wrong posture, uneven balance, or apply more pressure without knowing. So, there is a need for a simple system that can help patients during daily rehabilitation.

This project proposes a smart crutch system combined with a mobile application. The system measures the weight applied on the crutch using a load sensor and gives immediate feedback using LEDs and buzzer. The mobile app is used to set user details like weight and recovery stage, and based on that, the system decides safe limits.

The idea is to make rehabilitation safer and easier, especially for home use.

## **2. Motivation – Why I Am Interested**

This idea came from my personal experience.

A few months ago, I had an accident and injured my thigh bone. During recovery, I had to use crutches. At that time, I didn't know how much weight I should put on my leg. Sometimes I was afraid to walk properly, and sometimes I might have applied more pressure without knowing.

This made me realize that normal crutches only give support but do not guide the patient. That is when I thought about building a system that can help patients understand how they are walking and whether they are doing it correctly.

## **3. Novelty – What Makes the Idea Different**

- The system is based on user data, not fixed values
- It supports different modes like:
  - Fracture recovery
  - Post-surgery
  - Elderly support
- The mobile app calculates safe weight based on:
  - User weight
  - Recovery stage
- The app sends this data to the crutch using Bluetooth
- The crutch checks real-time weight and gives feedback
- It can also be extended with a motion sensor to detect wrong walking
- The app shows basic progress like current weight and alerts

This makes the system more useful compared to normal smart crutches.

## 4. Objectives – Goals of the Project

- To design a smart crutch system for rehabilitation
- To measure the weight applied by the user using a load sensor
- To compare the measured weight with safe limits
- To give immediate feedback using LEDs and buzzer
- To create a mobile app for entering user details
- To calculate safe weight based on recovery stage
- To send data from app to hardware using Bluetooth
- To improve patient awareness during walking
- To make the system simple, portable, and low-cost

## 5. Possible Approach – Methodology

The proposed smart rehabilitation crutch system measures the weight applied by the patient during walking using a load cell sensor. The signal from the sensor is amplified using the HX711 module and processed by a microcontroller. Based on the measured load, the system provides feedback through an LCD display, LED indicators, and a buzzer alert if the weight exceeds a safe rehabilitation limit. The data is also transmitted through a Bluetooth module to a mobile application for real-time monitoring and rehabilitation analysis.

### Step 1: User Input in App

The user enters details like weight and recovery stage in the mobile app.

### Step 2: Mode Selection

The app selects or allows the user to choose a mode:

- Fracture
- Post-surgery
- Elderly
- 

### Step 3: Weight Limit Calculation

The app calculates how much weight is safe based on the user's body weight and recovery stage.

### Step 4: Data Transfer

This value is sent to the crutch using Bluetooth.

### Step 5: Load Measurement

A load cell is placed in the crutch to measure the weight applied while walking.

### Step 6: Processing

The sensor value is amplified using HX711 and sent to the microcontroller.

### Step 7: Comparison

The microcontroller compares current weight with the allowed weight.

### Step 8: Feedback

- Safe → Green LED
- Moderate → Yellow LED
- Overload → Red LED + buzzer
- Step 9 (Optional): Gait Detection

Using a motion sensor, the system can detect imbalance or wrong walking.

### Step 10: App Monitoring

The app shows current weight, alerts, and simple progress tracking

## 6. Proposed System Architecture (Block Diagram)

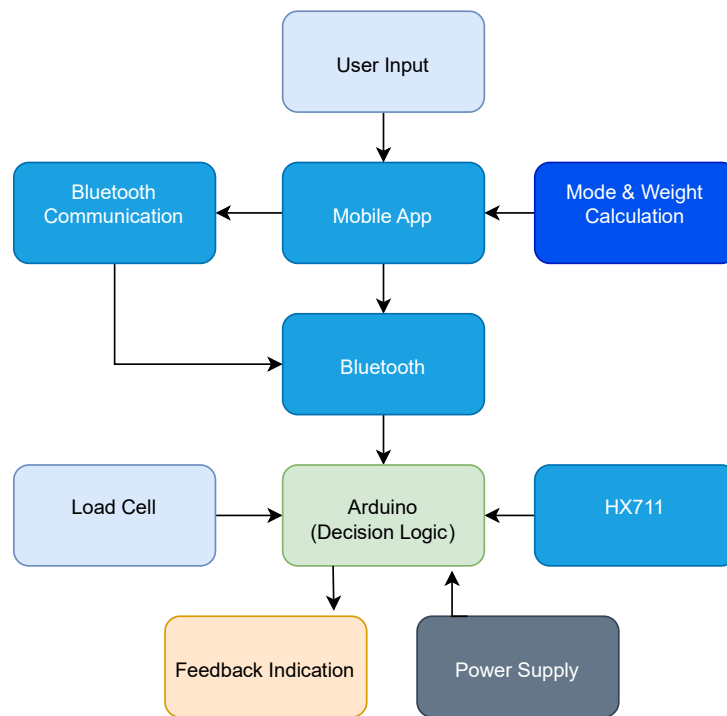


Fig.1: Block Diagram of Smart Rehabilitation Crutch

- **User Data Input**  
The user enters basic details such as body weight and recovery stage in the mobile application.
- **Mode Selection**  
Based on the input, the system selects or allows the user to choose a rehabilitation mode such as fracture, post-surgery, or elderly mode.
- **Safe Weight Limit Setting**  
The mobile application calculates the safe weight limit based on the user's body weight and recovery stage.
- **Bluetooth Communication**  
The calculated weight limit and selected mode are sent from the mobile application to the microcontroller using Bluetooth.
- **Load Sensing**  
A load cell placed in the crutch measures the real-time weight applied by the user while walking.
- **Signal Processing**  
The load cell output is amplified using the HX711 module and converted into digital data.
- **Microcontroller Processing**  
The microcontroller receives sensor data and compares the current weight with the allowed weight limit.
- **Decision Making**  
Based on comparison:
  - If within limit → safe condition
  - If near limit → caution
  - If above limit → unsafe condition

- **Feedback System**

The system provides immediate feedback:

- Green LED for safe
- Yellow LED for moderate
- Red LED and buzzer for overload

- **Optional Gait Monitoring**

A motion sensor can be used to detect imbalance, tilt, or incorrect walking patterns.

- **Mobile App Monitoring**

The application displays real-time data such as current load, alerts, and basic progress.

- **Future Expansion**

The system can be extended with cloud storage and remote monitoring for doctors.

## 7. Expected Output

- A working smart crutch prototype
- Real-time weight monitoring
- Immediate alert system
- Personalized safe weight limits
- Basic mobile app support
- Improved safety during rehabilitation

## 8. Future Scope

- Add cloud storage for data tracking
- Doctor monitoring system
- More detailed mobile dashboard
- AI-based prediction (in future)
- Advanced gait analysis